



Understanding Snapshots

As we move towards a world where data will drive everything, protecting this data is one of our highest priorities. From the primitive need of just having another copy of data just in case the hard disk crashes, to real-time data protection with near-zero RTO and RPO, data protection technology has made tremendous progress in the last 20 years or so. Even so, almost all methods still use snapshots as a basic tool. This write-up attempts to understand how snapshots work.

As the name suggests, snapshots are just a snap of the file system/volume/disk or, in other words, a point-in-time image. But what exactly is a point-in-time image? Let's get a little deeper into understanding what snapshots really are, how they work and, more importantly, what makes them an indispensable feature in data protection technology. A snapshot can be that of a disk, a volume, a file system, or even of a single file. So, if a disk is being snapped, its snapshot will be on another disk. Similarly, the snapshot of a file system is another file system, and of a file is another file, as we see in case of *vmdk* files in VMware. However, the underlying concept of a snapshot is similar.

As memory is divided into pages to address and locate data in it, the smallest unit of a data storage program is a logical block. Every file system divides the disk space into logical units of blocks of a size specified during creation. Block size usually depends on the type of workload. For instance, workloads that have larger files like databases use large size blocks. Every file is a collection of such blocks. When a new file is created, it is allocated an *inode*, which stores the first block address from where the file begins and the total number of blocks the file is allocated. Each time new data is written to the file, a number of

```
/dev/sda1      9.6G   4.4G   5.2G   46%
# tune2fs -l /dev/sda1 | grep -i 'block size'
Block size:      4096
```

Figure 1: Block size of an Ext4 file system on Ubuntu

blocks are allocated to it by the file system program. Figure 1 displays the block size of 4K of an *Ext4* file system on Ubuntu.

A file system basically consists of a superblock, journal, bitmaps and data blocks. The bulk of data is in data blocks. When a file system snapshot is triggered, all its blocks (excluding the data blocks) are copied to the snapshot. The data blocks of the original file system are in a 1:1 mapping with the data blocks of the snapshot. What this means is that the actual data continues to reside on the original file system. As seen in Figure 2, the snapshot only points to the same data blocks on the file system. So, when a backup application requests to read a particular file, the snapshot locates the starting data block of the file, reads the block on the original file system, and returns the data to the application. Since there is no actual data that is being read and written, snapshots are almost instantaneous. All the time they take is just to copy the metadata to know file allocations.

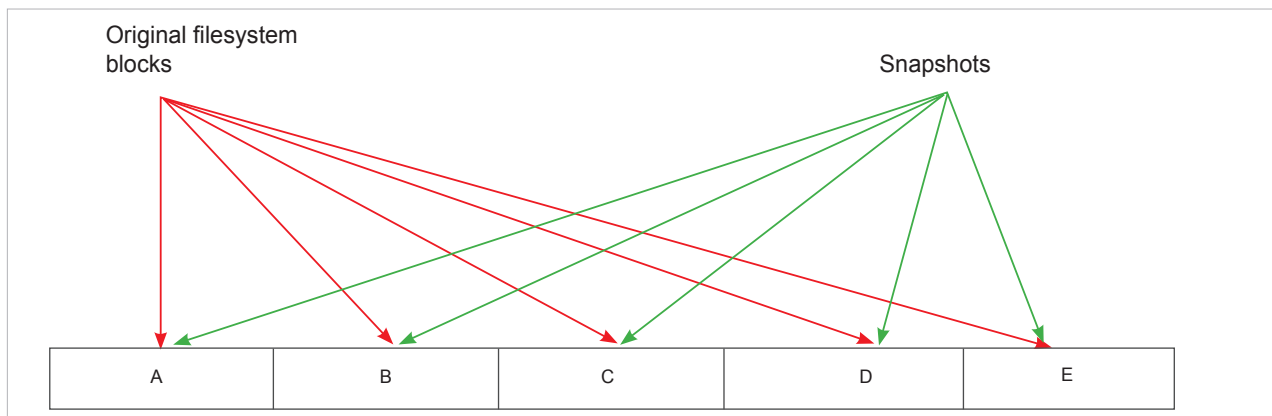


Figure 2: File systems and snapshots access the same disk blocks